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Patterns of Care and Outcomes of Proton and Eye Plaque Brachytherapy for Uveal Melanoma: Review of the NCDB Database



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Purpose/Objective(s): Radiotherapy for organ preservation has increased in popularity over enucleation in the modern management of choroid melanoma (CM). However, it is unknown whether eye plaque brachytherapy or proton external beam radiotherapy have equivalent outcomes.

Materials/Methods: Demographic and clinical data for 1224 patients with histologically confirmed choroid melanoma treated with either brachytherapy or proton beam therapy from 2004-2013 were obtained from the National Cancer Database (NCDB). Patients were excluded if tumor thickness and basal diameter were not recorded. Patients were staged T1-4N0M0 from the recorded tumor dimensions according to AJCC 7th edition. Logistic regression and propensity score matching was used to create a 1:1 matched cohort. Kaplan-Meier and Cox regression analyses were performed to evaluate survival in brachytherapy and proton groups with and without propensity score matching. Log rank was used to compare survival.

Results: Median follow up was 37 and 29 months for brachytherapy and protons, respectively, and 93% of patients were treated from 2010-2013. The majority of patients were treated with brachytherapy (n = 996) vs. protons (n = 228). There were no differences in age (61 years), sex, insurance status, Charlson-Deyo score, T stage, ciliary/extraocular extension, or tumor diameter or thickness. Patient receiving protons were more likely to be treated at an academic center (99 vs. 76%), non-white (86 vs. 96%), travelled >60 miles to treatment (81 vs. 72%), started treatment > 60 days from diagnosis (19 vs 11%); Proton patients also came from more urban, affluent and educated zip codes (all p < 0.004). Median proton dose was 56 Gy in 4 fractions. Two year overall survival was 96% vs. 92%, and 5 year overall survival was 81 vs. 54% for brachytherapy and protons, respectively (p < 0.001). Cox regression found older age (HR = 1.05), larger tumor diameter (12-18mm, HR = 2.25, >18mm, HR = 3.56), academic facility (HR = 2.07), and protons (HR = 1.91) were negative prognosticators of survival (all p < 0.02). In the propensity-score matched cohort, two year overall survival was 97% vs. 93%, and 5 year overall survival was 77% vs. 51% for brachytherapy and protons, respectively (p = 0.008). Cox regression found older age (HR = 1.06), larger tumor diameter (12-18mm, HR = 2.48, >18mm, HR = 6.41), and protons (HR = 1.89) were negative prognosticators of survival (all p < 0.02).

Conclusion: Treatment with protons is associated with inferior survival outcomes compared to brachytherapy in this retrospective analysis. There

may be unaccounted variables not reported to the NCDB that influence survival, necessitating further prospective research.

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Daily Localization Impact on Decision of Split-Field or Whole-Field IMRT for Head and Neck Cancer Radiation Treatment



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Purpose/Objective(s): Radiation of the head and neck cancer can be accomplished using two methods: split-field IMRT (SFI), in which the LAN is treated with a separate anterior field and the extended whole-field IMRT/VMAT (WFI) in which LAN is included with the IMRT/VMAT field. Larynx can be spared more extensively by SFI; however, with daily localization uncertainties, larynx could receive extra dose at the matching line. This study is to determine if SFI has the advantage of saving larynx throughout patient treatment delivery.

Materials/Methods: Nine common head and neck cancer cases were analyzed. Both SFI and WFI at upper target coverage were generated with VMAT. For each case, the same planning criteria were applied to the target: 70 Gy (high risk) 60 Gy (intermediate risk) with 63Gy and 56Gy secondary target coverage; therefore, similar target coverage and dose falloff can be observed in both techniques. Larynx sparing can also be observed through its mean dose. Thirty days of kV cone beam CT (CBCT) images on each case were also delineated with contralateral and ipsilateral target as well as larynx as critical structure. About 300 CBCT images with daily delivered doses were analyzed and compared in a form of dose-volume histograms.

Results: Averaged dose for Larynx with WFI is 37.0Gy as planned with simulation CT and that for SFI is 17.4Gy. Normalized daily larynx doses to the planned doses for WFI and SFI indicate that SFI with daily localization uncertainties results in 10% higher in larynx mean dose which is lower than which with WFI, 15%; however, the SFI can have much higher hot spots in the larynx and much lower cold spots in target than WFI does.

Conclusion: Patients treated with SFI indeed shows more sparing at the treatment planning than those with WFI with similar target coverage; however, lower cold spots in target and higher hot spots can be observed in these 300 CBCT images based dose distributions for these nine patients. These cold spots in targets could contribute to treatment failure and hot spots in larynx can be detriment to the organ function.

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WFI	Volume averaged (c.c)	Vol sd	Averaged Min Dose (Gy)	Ave. min SD	Averaged Mean Dose (Gy)	Ave. Mean sd	Averaged Max Dose (Gy)	Ave. Max SD
Contralateral target	64.49	2.25	31.79	11.32	56.77	3.93	63.63	2.05
Ipsilateral target	80.63	5.41	35.95	9.01	67.48	3.41	76.87	0.85
Larynx	43.52	1.47	21.65	11.98	42.48	6.91	75.28	1.61
SFI	Volume averaged (c.c)	Vol sd	Averaged Min Dose (Gy)	Ave. min SD	Averaged Mean Dose (Gy)	Ave. Mean sd	Averaged Max Dose (Gy)	Ave. Max SD
Contralateral target	58.59	1.81	14.14	10.62	51.06	2.46	59.24	1.75
Ipsilateral target	59.34	1.32	8.25	5.34	54.91	3.19	68.39	4.01
Larynx	40.16	1.56	5.41	0.48	18.93	2.38	62.97	5.91